VALIDATION

TESTING FRAMEWORK.

1,000+ HOURS VALIDATED.

Comprehensive testing methodology and results for Tobe Energy's electrolyzer system.

>92%	8×	<30°C	0
SYSTEM EFFICIENCY	LESS HEAT LOSS	OPERATING TEMP	RARE EARTH METALS

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EFFICIENCY TESTING FRAMEWORK

Discover how Tobe Energy is redefining hydrogen production through rigorous efficiency testing and cutting-edge instrumentation.

This document provides a detailed analysis of Tobe Energy's methodology for measuring and enhancing the efficiency of its novel hydrogen production system. Through advanced instrumentation, real-time data acquisition, and robust validation techniques, we ensure precision, transparency, and repeatability in efficiency calculations. By showcasing our results and scaling plans, this report highlights Tobe Energy's commitment to delivering economically viable and environmentally sustainable hydrogen solutions.

OPTIMIZING HYDROGEN PRODUCTION: A COMPREHENSIVE EFFICIENCY TESTING FRAMEWORK

In the evolving landscape of renewable energy, hydrogen production plays a pivotal role in providing a clean, sustainable alternative to fossil fuels. The efficiency of hydrogen production systems is critical, as it directly impacts both the economic viability and environmental sustainability of hydrogen as a fuel source.

At Tobe Energy, we have developed a novel electrolysis-based hydrogen production system designed to maximize efficiency by reducing energy losses, improving operational reliability, and minimizing waste. This paper outlines the methodology used to measure the efficiency of our hydrogen production system, detailing the instrumentation and calculations employed, and reviews results of some of our efficiency testing.

By carefully selecting and integrating sensors, we have developed a comprehensive testing framework that accurately captures the key parameters affecting system performance. This approach ensures that the data collected is not only precise but also easily replicable, allowing for consistent monitoring and improvement over time.

The goal of this white paper is to provide a clear, step-by-step guide on how our system's efficiency is calculated, from measuring power input to quantifying hydrogen output. Each piece of instrumentation used, along with its unique tag number, will be detailed in the following sections, along with how the collected data feeds into the overall efficiency calculation.

This methodology is designed to ensure transparency and repeatability, making it a valuable tool for internal testing, external validation, and future system optimization. By laying out this process, we aim to showcase the rigorous testing and validation procedures that underpin the performance of our hydrogen production system.

As efficiency remains one of the most important metrics in evaluating any energy system, our methodology reflects our commitment to developing a solution that is both economically viable and environmentally responsible.

CIRCUIT BLOCK FLOW DIAGRAM OVERVIEW

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The Circuit Block Flow Diagram provides a high-level representation of the electrical components in the hydrogen production system. This diagram outlines the flow of electrical power through the system, starting from the power supply and flowing through various sensors and circuits before reaching the electrolysis unit.

Key elements such as power sensors (PoT-01A/B, PoT-02, PoT-03) and current clamps (AG-01) are annotated, allowing for real-time monitoring of power consumption and input levels. Additionally, the diagram showcases the integration of the signal processing and amplification units, ensuring that electrical power is optimized for efficient hydrogen production.

The layout of the circuit is designed to provide precise feedback on voltage, current, and power at each critical stage, enabling accurate calculations of the total energy input into the system. By incorporating both real-time monitoring and secondary validation points, the Circuit Block Flow Diagram ensures transparency in the measurement of power consumption.

SIMPLIFIED PIPING & INSTRUMENTATION DIAGRAM

The Piping & Instrumentation Diagram (P&ID) outlines the physical and chemical processes involved in the hydrogen production system. It provides a detailed look at the flow of fluids, gases, and electrolytes throughout the system, from the water supply tank to the electrolysis cell and the gas storage units.

Critical instrumentation, such as flow transmitters (FT-E4, FT-P1), pressure sensors (PT-E3, PT-B6), and level sensors (LT-T2, LT-E1, LT-B4), are strategically placed to ensure continuous monitoring of the system's operational conditions.

The diagram also illustrates the flow of oxygen and hydrogen gases through the system, highlighting key components like the bubbler, where the oxygen concentration is measured (OT-B3), and the hydrogen sensor (HT-B2), which ensures safe operation and efficient gas separation.

Each measurement point is designed to feed real-time data into the control system, enabling the precise tracking of gas production and water consumption. The Process Flow Diagram ensures that the overall system remains balanced and that gas production rates can be optimized for maximum efficiency.

INTRODUCTION TO EFFICIENCY CALCULATION

The efficiency of our hydrogen production system is determined by comparing the total energy input to the system with the energy content of the hydrogen produced. Accurately measuring these values requires a comprehensive instrumentation setup, ensuring that every relevant parameter — from electrical power to gas flow and composition — is captured in real-time.

By leveraging a combination of sensors and validation instruments, we are able to achieve precise control over the data collection process, allowing for reliable and reproducible efficiency calculations. The key components contributing to the calculation include:

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1. **Total Power Input**
2. **Gas Production Measurement**
3. **Gas Production & Composition Validation**

When demonstrating calculations throughout this paper, the tag numbers of each instrument will be referenced as variables. This approach streamlines the calculations, making it easier to trace how each measurement is integrated into the efficiency formula, while maintaining consistency with the system's instrumentation and diagrams.

The calculations outlined in the following sections detail how each of these measurements is used to compute the overall efficiency of the system. Our approach accounts for both electrical and gas-related variables, allowing us to present a complete picture of the system's energy conversion effectiveness.

TOTAL POWER INPUT

Measured using high-precision power sensors, the total electrical energy supplied to the electrolysis system is carefully monitored. Secondary verification tools — including current clamps and analog meters — are used to validate these readings.

The total power input to the system is calculated from the sum of measured inputs across all power sensors. To calculate the switching losses, the difference between input power (PoT-01A/B) and output power (PoT-02) is computed. Both values are in watts.

GAS PRODUCTION MEASUREMENT

The volumetric flow rate of the gas produced, primarily hydrogen, is tracked through flow sensors, and corrections are made for relative humidity and water vapor content. By accounting for these factors, we

ensure the measured gas volumes accurately reflect the dry gas output of hydrogen and oxygen.

Flow Rate Measurement

The volumetric flow rate of gas is measured by flow sensor FT-E4, positioned at the electrolysis cell outlet. Corrections are applied for relative humidity and water vapor content to ensure the measured gas volumes accurately reflect dry hydrogen and oxygen output.

Relative Humidity Correction

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The gas exiting the electrolysis cell is typically saturated with water vapor; accordingly, the measured gas flow is corrected for relative humidity. A temperature-corrected humidity sensor, RHT-B1, positioned in the bubbler, provides real-time data on the relative humidity of the gas. This allows us to determine the fraction of the gas flow attributable to water vapor, which is subtracted from the total flow to yield the dry gas flow rate.

To mitigate the possibility of air present in the bubbler prior to system testing, the bubbler water is started full and manually drained to reduce the level once gas production has begun. In the following formulas, the variable P represents pressure, distinct from P representing power in the previous formulas.

The partial pressure of water vapor is calculated as:

> **P_H2O = RH (RHT-B1) × P_SAT**

Where:

- **P_H2O** is the partial pressure of the water vapor
- **RH (RHT-B1)** is the relative humidity measured by the sensor (values 0.0–1.0)
- **P_SAT** is the saturation vapor pressure of water at gas temperature (sourced from a reference table or an Excel macro)

With the partial pressure of water vapor established, the corrected volumetric gas flow is:

> **Q_DRY = Q_MEASURED (FT-E4) × (P_TOTAL (PT-E3) − P_H2O) / P_TOTAL (PT-E3)**

Where:

- **Q_MEASURED (FT-E4)** is the volumetric flow measurement
- **P_TOTAL (PT-E3)** is the pressure of gas produced in the electrolysis cell
- **P_H2O** was calculated above

Finally, per the stoichiometric ratio, the dry gas is **two-thirds hydrogen by volume**. The volumetric quantity of hydrogen gas produced, corrected for relative humidity, is:

> **Q_H2 = (2/3) × Q_DRY** (liters per minute of hydrogen produced)

GAS PRODUCTION & COMPOSITION VALIDATION

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While the calculations above are straightforward and verifiable through published literature — and the sensors are reinforced with analog gauges and meters for cross-checking — several additional layers of validation have been implemented.

For gas composition, an oxygen sensor (OT-B3) measures oxygen concentration in the bubbler with **±1% accuracy** across a **0–100% range**. Hydrogen concentration is measured up to **10,000 ppm** using a dedicated hydrogen sensor (HT-B2).

To accurately track the buildup of hydrogen in the system, testing begins with air in the bubbler and hydrogen levels are monitored as they rise to 10,000 ppm. When this sensor is not actively verifying hydrogen concentration in startup systems, it serves as a leak detection instrument.

Furthermore, by tracking the flow rate of water through the dosing pump and monitoring water levels across all vessels — from the water tank to the electrolysis cell and the bubbler — a precise mass balance of the system is maintained. Temperature and pressure measurements in both the electrolysis cell and the bubbler are factored into these calculations, ensuring that all "lost" mass corresponds to the volume of gas produced.

The pH of the system is also tracked, which is particularly important for efficiency tests that simulate sea brine conditions. pH monitoring serves as an additional verification check against an analog TDS meter, providing a further layer of accuracy in efficiency testing.

PUBLISHED RESULTS FROM SBIR GRANT TESTING

Our efficiency testing has been validated through a series of experiments conducted as part of our SBIR grant. These tests were carried out using both low and high TDS water, following the exact methodology outlined in this paper, with one notable exception: total power input was measured using a wall outlet plug power meter positioned between the 120VAC power source and our Siglent DC Power Supply.

The reported efficiency values inherently include losses from the DC power supply, which is designed for versatility rather than maximum efficiency. The Siglent power supply used in testing operates at an efficiency of approximately **79.1% to 79.8%**, as detailed in the manufacturer's efficiency report.

Excluding the losses attributable to the laboratory power supply — by adding the average observed efficiency difference of **20.2% to 20.9%** to the reported values — results in adjusted system efficiencies ranging from **89.6% to 92.8% LHV**.

SCALING OF HYDROGEN PRODUCTION

Testing covered two system configurations:

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- **1 kW Unit**: Designed specifically for high TDS contaminant testing.
- **2.5 kW Flagship Prototype**: Capable of producing more than **5 kg H_2/day**.

For the 2.5 kW low TDS configuration, hydrogen mass produced — standardized over **24 hours of continuous operation at full capacity** — amounted to **7.7 kg**. This demonstrates that hydrogen production scales linearly with power input and anode+cathode surface area as system size increases.

Preliminary data indicate confidence in scaling this process to **25 kg H_2/day** with no significant loss of efficiency. The system is believed to be scalable beyond this threshold given sufficient time and resources for further testing.

FUTURE RESULTS AND BATTERY-POWERED OPERATION

In addition to the data published here, we have explored powering the flagship prototype using more efficient AC/DC conversion methods, batteries, and solar arrays. While the detailed data for these tests have not yet been released, system efficiencies between **90% and 95% LHV** have been observed for hydrogen production exceeding **5 kg/day**.

These results will be included in a forthcoming report. We can currently confirm an average efficiency of **94.7% LHV** observed over a 10-hour test of our flagship **5 kg/day prototype** — representing stack-level efficiency. At the system level, including all balance-of-plant auxiliaries, efficiency exceeds **92% HHV**, corresponding to a system energy consumption of approximately **~42 kWh/kg**.

INSTRUMENTATION AND SYSTEM CONNECTIVITY

Overview of Instrumentation

Accurate measurement is fundamental to determining system efficiency. We have carefully selected and integrated a range of sensors throughout our hydrogen production system. Each sensor is tagged and

mapped to either the Process Diagram (P&ID) or the Circuit Diagram, ensuring clarity and traceability during testing.

The primary role of these instruments is to capture real-time data related to power input, hydrogen output, and system conditions — all of which feed into our efficiency calculations.

Integration and Monitoring with Arduino Instrumentation

The entire instrumentation setup is connected via an Arduino-based system, which acts as the central hub for sensor data collection. The sensors are programmed to interface with the Arduino platform, enabling seamless data acquisition.

To enable remote monitoring and enhance system visibility, a custom dashboard has been built using the Arduino Cloud platform. This dashboard allows real-time monitoring of system parameters — including voltage, current, temperature, and hydrogen flow rate — making the efficiency testing process highly accessible and transparent.

LIST OF INSTRUMENTS AND SPECIFICATIONS

For each instrument below, we provide specifications, the corresponding tag number from either the Process Diagram or Circuit Diagram, and a brief description of what it measures.

Power Sensor

- **Tag Numbers**: PoT-01A/B, PoT-02, PoT-03 (Block Flow Circuit Diagram)
- **Sensor Model**: INA260
- **Specifications**:
  - Voltage Range: 0–36V DC
  - Current Range: ±15A
  - Power Range: Up to 540W (calculated from voltage and current ranges)
  - Accuracy: **±0.02%** for voltage, **±0.1%** for current
- **Measurement**: Monitors voltage, current, and power in real-time.
- **Purpose**: The INA260 monitors total electrical power input into the electrolysis system, combining voltage and current readings to calculate power and ensure accurate tracking of energy supplied.
- **Integration**: I2C interface for real-time data acquisition.
- **Note**: Since voltages at PoT-03 exceed the INA260's measurement limits, a voltage divider circuit steps down the voltage. The Arduino code references the divider's characteristics to return correct voltage values, ensuring safe and accurate measurement of high-voltage signals.

Temperature Sensor

- **Tag Numbers**: TT-E2, TT-B5 (Simplified P&ID)
- **Sensor Model**: DS18B20
- **Specifications**:
  - Temperature Range: −55°C to +125°C
  - Accuracy: **±0.5°C** from −10°C to +85°C
  - Resolution: Programmable (9 to 12-bit)

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- **Measurement**: Measures the temperature of the electrolyte within the electrolysis cell.
- **Purpose**: Monitors electrolyte temperature to track heat losses and thermal behavior, both of which are important for analyzing system efficiency.
- **Integration**: Connected to the Arduino via a One-Wire interface, allowing multiple sensors on the same data line.

Environmental Reference Sensor

- **Tag Number**: RefT-R1 (Simplified P&ID)
- **Sensor Model**: BME280
- **Specifications**:
  - Temperature Range: −40°C to +85°C
  - Humidity Range: 0% to 100% RH
  - Pressure Range: 300 hPa to 1,100 hPa
- **Accuracy**:
  - Temperature: **±1.0°C**
  - Humidity: **±3%**
  - Pressure: **±1 hPa**
- **Measurement**: Measures ambient temperature, humidity, and barometric pressure.
- **Purpose**: Captures environmental conditions during testing. Ambient temperature, pressure, and humidity can affect hydrogen production, and this data helps adjust and contextualize efficiency calculations.
- **Integration**: Connected to the Arduino via I2C interface.
- **Note**: The BME280 provides critical environmental data for correlating system performance with external conditions.

Pressure Sensor

- **Tag Number**: PT-E3, PT-B6 (Simplified P&ID)
- **Sensor Model**: DFRobot Sen0257
- **Specifications**:
  - Pressure Range: 0–232 psig (16 bar)
  - Accuracy: **±0.5%** of full scale
  - Interface: Analog (0.5V to 4.5V output)

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- **Measurement**: Measures the pressure of gases or liquids in the system.
- **Purpose**:
  - PT-E3: Monitors pressure at the electrolysis cell to ensure stable operating conditions.
  - PT-B6: Monitors pressure at the bubbler to verify safe and consistent gas flow.
- **Integration**: Connected to the Arduino via analog input pins; voltage output is converted to pressure using a calibration curve.

Level Sensors

- **Tag Numbers**: LT-T2, LT-E1, LT-B4 (Simplified P&ID)
- **Sensor Model**: MaxBotix HRLV-EZ1
- **Specifications**:
  - Measurement Range: 1 mm to 500 cm
  - Accuracy: **±1%** of detected range
  - Resolution: 1 mm
  - Output Type: Pulse width, analog voltage, or serial output
- **Measurement**: Measures distance from sensor to liquid surface, converted to liquid level.
- **Purpose**:
  - LT-T2: Monitors liquid level in the water tank to ensure proper filling and prevent overflow.
  - LT-E1: Tracks water level within the electrolysis cell for operational stability.
  - LT-B4: Monitors bubbler water level to maintain safe operations.
- **Integration**: Connected to the Arduino via pulse width, analog, or serial output, allowing flexible and precise liquid level measurement.
- **Note**: The HRLV-EZ1 provides enhanced accuracy and noise filtering, ensuring stable, continuous readings suitable for real-world operational environments.

Flow Transmitter

- **Tag Number**: FT-P1 (Simplified P&ID)
- **Sensor Model**: DFRobot Gravity Liquid Flow Sensor
- **Specifications**:
  - Flow Range: 0.15–1.5 L/min
  - Accuracy: **±3%**
  - Output Type: Digital pulse output

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- **Measurement**: Measures the flow rate of liquid passing through the Atlas Scientific Dosing Pump.
- **Purpose**: FT-P1 monitors the real-time flow rate of the dosing pump, ensuring the correct volume of liquid is delivered during operation — critical for accurate dosing and flow control.
- **Integration**: Connected to the Arduino via a digital input pin; pulse output is used to calculate flow rate by counting pulses per liter, providing real-time readings between 0.15 and 1.5 L/min.

Gas Flow Transmitter

- **Tag Number**: FT-E4 (Simplified P&ID)
- **Sensor Model**: FS1015
- **Specifications**:
  - Flow Range: 0–50 L/min (air or non-corrosive gases)
  - Accuracy: **±2%**
  - Output Type: Analog voltage output
- **Measurement**: Measures the flow rate of gas produced in the electrolysis cell.
- **Purpose**: FT-E4 monitors the flow of gas exiting the electrolysis cell, providing real-time data on gas flow rates.
- **Integration**: Connected to the Arduino via an analog input pin; the sensor's analog output is converted to liters per minute using calibration data in the Arduino code.

pH Transmitter

- **Tag Number**: PhT-T1, PhT-B7 (Simplified P&ID)
- **Sensor Model**: Atlas Scientific pH Meter
- **Specifications**:
  - pH Range: 0–14
  - Accuracy: **±0.002 pH**
  - Output Type: Analog or I2C (configuration-dependent)

- **Measurement**: Measures the pH level of water in the system.
- **Purpose**:
  - PhT-T1: Monitors pH of water in the tank, ensuring it remains within the optimal range for efficient electrolysis.
  - PhT-B7: Monitors pH in the bubbler to maintain stable operating conditions and detect chemical balance shifts.
- **Integration**: Atlas Scientific pH sensors are connected to the Arduino via I2C interface for real-time data acquisition.

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Humidity and Temperature Transmitter

- **Tag Number**: RHT-B1 (Simplified P&ID)
- **Sensor Model**: DHT22
- **Specifications**:
  - Humidity Range: 0% to 100% RH
  - Accuracy (Humidity): **±2% RH**
  - Temperature Range: −40°C to +80°C
  - Accuracy (Temperature): **±0.5°C**
  - Output Type: Digital (single-wire)
- **Measurement**: Measures relative humidity and ambient temperature in the system environment.
- **Purpose**: RHT-B1 monitors ambient humidity in the vapor space of the bubbler to ensure accurate hydrogen and oxygen production calculations.
- **Integration**: Connected to the Arduino via a single-wire digital interface for real-time humidity and temperature monitoring.

Hydrogen Leak Detection Sensor

- **Tag Number**: HT-B2 (Simplified P&ID)
- **Sensor Model**: Figaro TGS821
- **Specifications**:
  - Hydrogen Detection Range: 500–10,000 ppm
  - Accuracy: **±30%** at 1,000 ppm H_2
  - Output Type: Analog voltage
- **Measurement**: Detects the presence of hydrogen in air to monitor potential leaks.

- **Purpose**: HT-B2 is primarily used to detect hydrogen leaks and verify that the system achieves a hydrogen capture rate of greater than **99%**. The sensor triggers an alert if hydrogen levels exceed safe thresholds, enabling immediate corrective action.
- **Integration**: Connected to the Arduino via an analog input pin; voltage output is monitored continuously to detect even minor hydrogen leaks.

Oxygen Sensor

- **Tag Number**: OT-B3 (Simplified P&ID)

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- **Sensor Model**: Figaro KE-25
- **Specifications**:
  - Oxygen Concentration Range: 0% to 100% O_2
  - Accuracy: **±1% O_2**
  - Output Type: Analog voltage output
- **Measurement**: Measures oxygen concentration in the bubbler.
- **Purpose**: OT-B3 monitors oxygen concentration in the bubbler, enabling derivation of remaining hydrogen concentration (since what is not oxygen is hydrogen). Combined with readings from the hydrogen sensor (HT-B2), this validates the efficiency of gas separation and confirms the system maintains the desired hydrogen concentration.
- **Integration**: Connected to the Arduino via an analog input pin, providing real-time oxygen concentration data for continuous gas composition monitoring.
- **Note**: This sensor plays a critical role in validating the hydrogen capture rate and ensuring safe, efficient operation by measuring oxygen levels as a proxy for hydrogen concentration.

ADDITIONAL VERIFICATION INSTRUMENTS

While primary data acquisition for efficiency calculations is handled through Arduino-compatible sensors, several standalone instruments are employed to cross-check and verify key measurements. These serve as secondary validation points, ensuring all collected data is accurate and consistent.

AC Current Measurement

- **Location**: AG-01 (Circuit Block Flow Diagram)
- **Instruments**:
  - AstroAI CM2KOR AC Clamp Meter: A standalone clamp meter for measuring alternating current in power supply lines.

    - Hantek CC-65 AC/DC Current Clamp: An oscilloscope probe capable of measuring both AC and DC currents, providing detailed waveform data.
- **Purpose**: These tools cross-check current measurements collected by the system's Arduino sensors, ensuring precision in power calculations.

Gas Flow Measurement

- **Location**: FG-E5 (Simplified P&ID)
- **Instrument**: Welding Gas Acrylic Flowmeter
- **Range**: 0–50 L/min
- **Purpose**: Verifies gas flow rates — particularly for hydrogen and non-corrosive gases — providing an additional manual check against Arduino-monitored flow rates.

Power Supply Analog Gauges

- **Location**: Resonant Tank Circuit
- **Instruments**: Analog voltage and current gauges mounted on the power supply cases.
    - These gauges provide real-time visual feedback on voltage and current levels within the Resonant Tank Circuit. They are used primarily for operational monitoring and can be toggled on or off, as their presence may affect system performance.
- **Purpose**: The analog gauges offer a quick manual reference for voltage and current conditions, serving as a secondary validation tool alongside digital measurements captured by the Arduino system.

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